ARDUINO

Lec4

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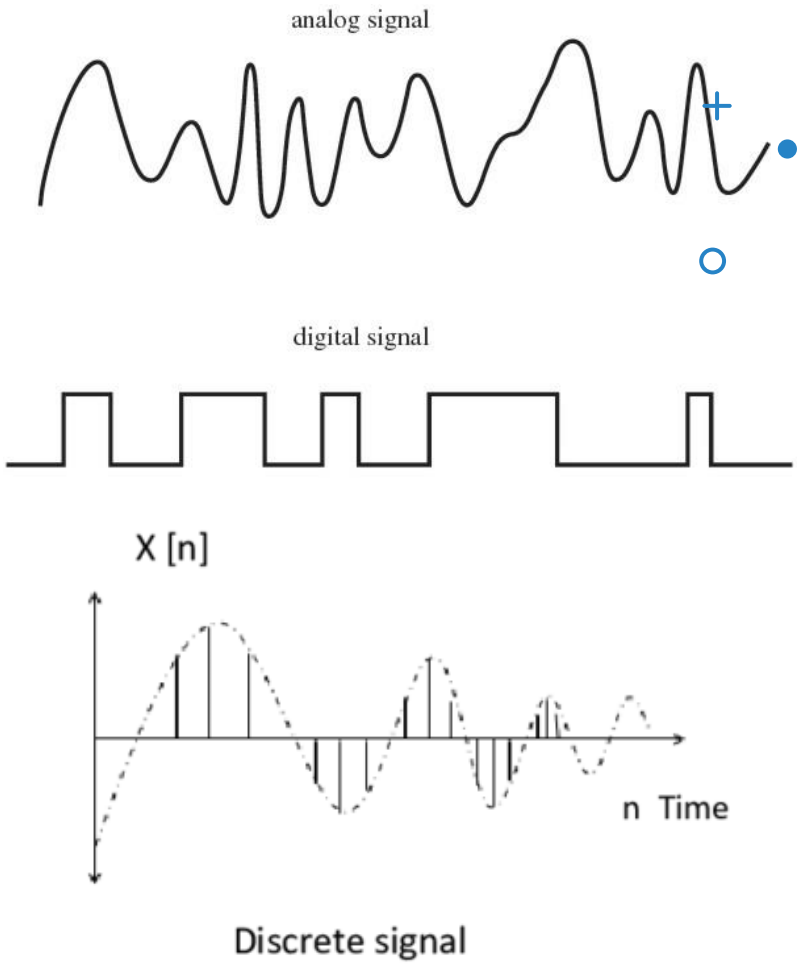
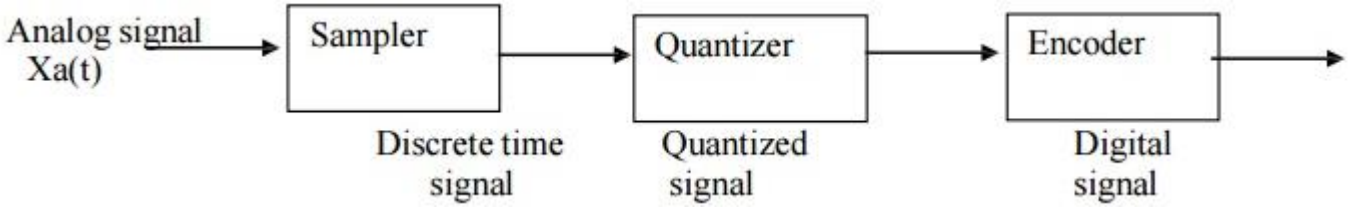
❑ Introduction

- There are 2 types of signals :
  1. Analog : is a signal which is can have any value (such as sin wave)
  2. Digital : is a signal which has only 2 values 0 or 1 .
- There are 2 types of converters for signal :
  1. ADC : Analog to digital converter .
  2. DAC : Digital to analog converter .
- There is a type of signals which is an intermediate between the analog to digital , and its name is discrete signal .
- We can get the discrete signal by sampling the wave in time with a sampling rate (frequency) .
- Sampling rate increases → the accuracy will i



❑ Discrete signal variables

- Sampling rate → the num which describe how the one value repeats .
- Resolution → is the least value you can get from the signal and less the value you get , better accuracy you get . (rate is inversely proportional to the resolution)
- Num of values in the discrete signal is the no. of bits (n).
- The no. of levels (L) = 2 ^ n .
- To convert from the analog to discrete we need to sample the analog signal , and to convert from the discrete to analog , we need a Quantization for the levels .



✓ Some equations

- Step (delta) = $\frac{max.voltage - minimum\ voltage\ (refrence)}{2^n\ (\ no\ .of\ levels\ )}$
- N (number of level) = $\frac{input\ voltage}{step\ (\ delta\ )}$
- Input voltage = N * step (delta)

- We have a lot of application such as temperature sensor or voltage reader .

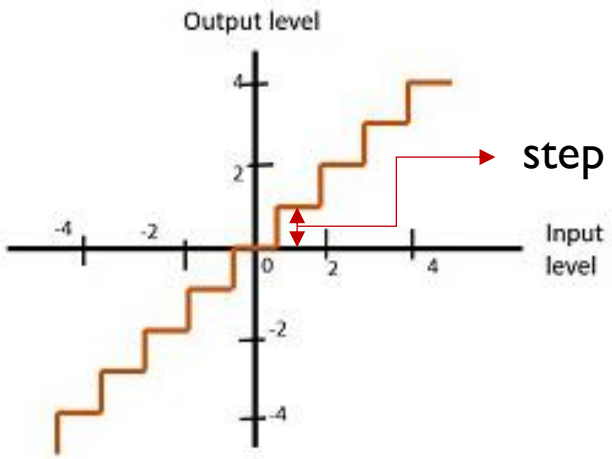


Fig 2 : Mid-Tread type Uniform Quantization

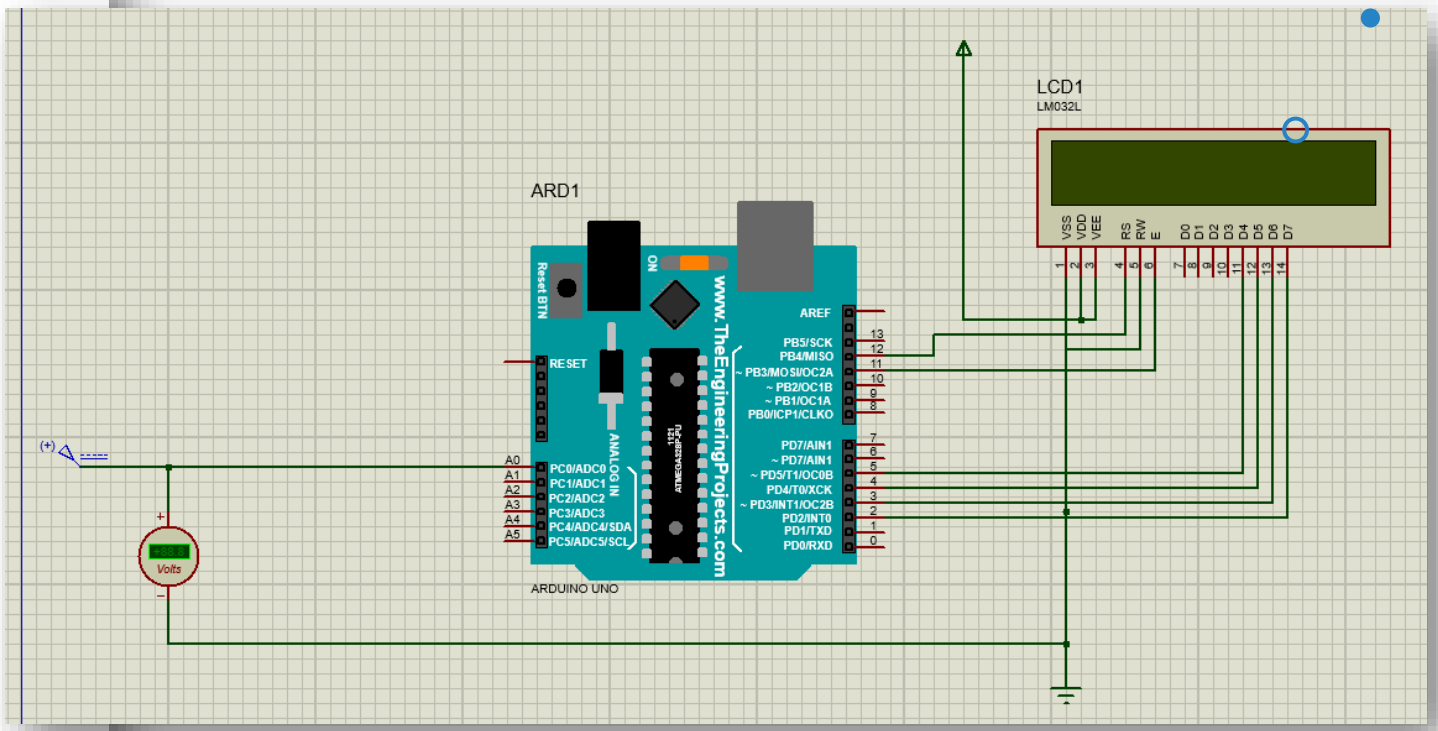
❖ Some Codes :

- ADC → To read the no. of the level of the data → analogRead(no. of input pin) .
- The first code returns → (0 : 1023) .
- DAC (PWM) → analogWrite(no. of output_pin , value) returns a value (0 : 255) .

❖ The calculations for DAC Or PWM in the slide of DAC .

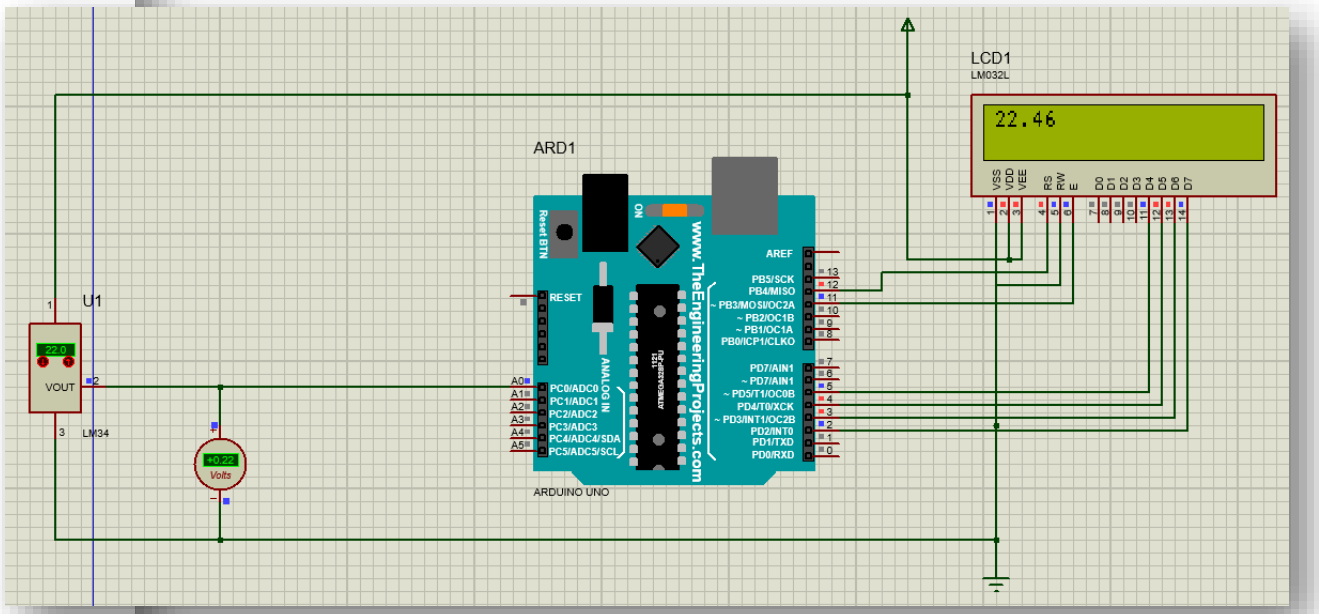
✓ EX :- voltage reader

```
# define V0 A0
#include <LiquidCrystal.h>

const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

void setup() {
  // put your setup code here, to run once:
  pinMode(A0,INPUT);
  lcd.begin(16,2);

}

float in_v;
int level ;

void loop() {
  // put your main code here, to run repeatedly:
  level = analogRead(V0);
  in_v= level * ( 5.0/1024 );
  lcd.print(in_v);
  delay(100);
  lcd.clear();
}
```



✓ EX :- Temperature reader

```
# define V0 A0
#include <LiquidCrystal.h>

const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

void setup() {
  // put your setup code here, to run once:
  pinMode(A0,INPUT);
  lcd.begin(16,2);

}

float in_v;
int level ;

void loop() {
  // put your main code here, to run repeatedly:
  level = analogRead(V0);
  in_v= level * ( 5.0/1024 );
  lcd.print(in_v*100);
  delay(100);
  lcd.clear();
}
```



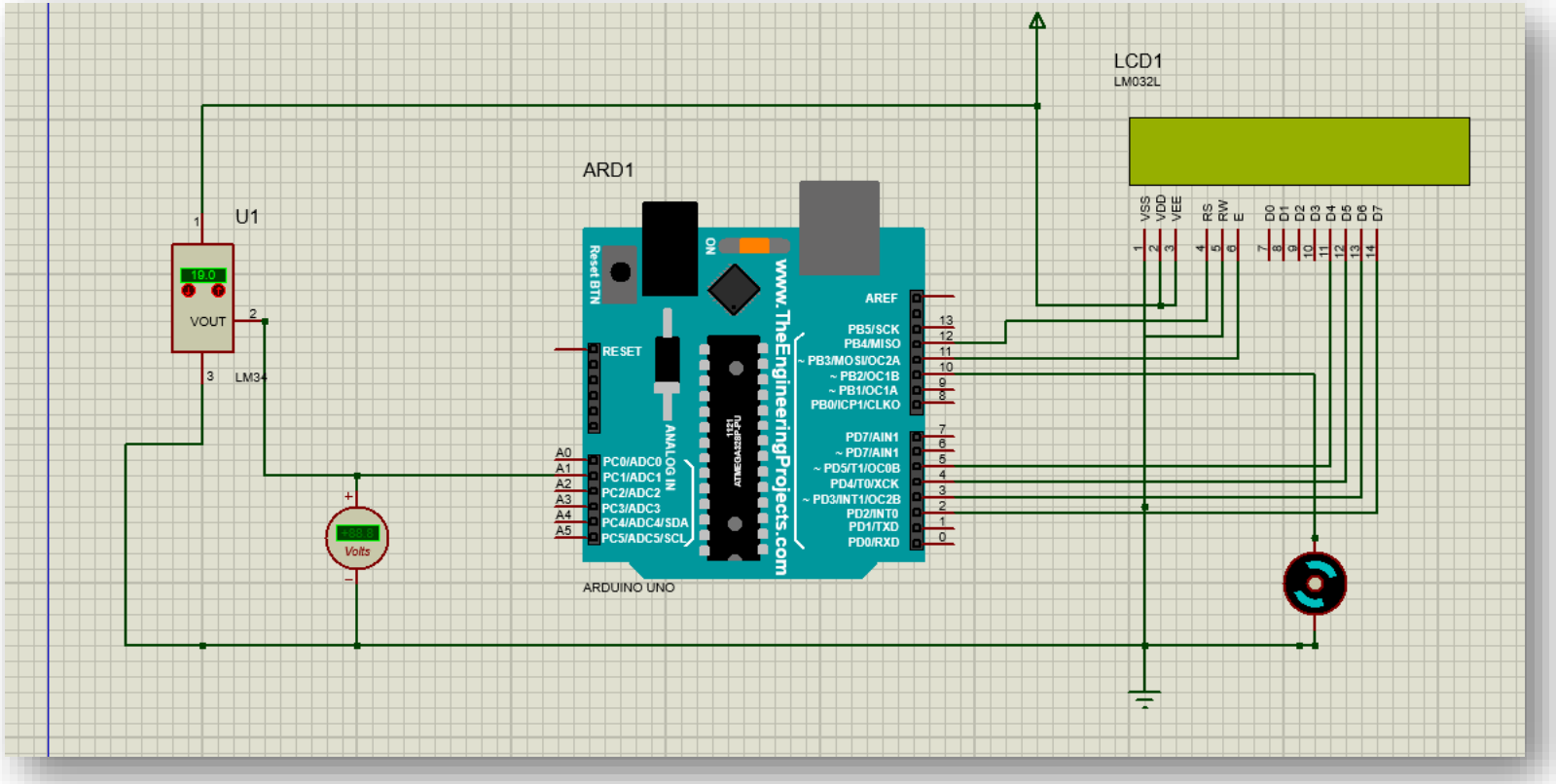
❑ DAC

- There is no DAC in ARDUINO but we have a part which perform the same task of the DAC .
- This part in ARDUINO is called PWM (Pulse Width Modulation) is marked with (~) in ARDUINO .
- We must know how we get the analog signal from the digital one by calculating the average of the digital signal → $\frac{1}{T}\int_0^T v(t)\ dt$
- The average voltage is equal to (the maximum voltage * (time of high output))
- We can use this average power for any purposes (for dc-motor) .
- Duty cycle is the ratio from the tot. signal in which the output is high .
- The analog writing is from 0 to 255 → the representation of high part = $\frac{v(\ req\ av.voltage\ )}{5} * 255$

❑ Applications of DAC

- We can use it to output a certain voltage to control the motor speed according to the temperature for example .
- More examples ….. .
- Here , we will perform the example of motor .

✓ Ex : - Controlling the speed of motor according to the temp ?



```
#define TEMP A1
#define MOTOR 10
#include <LiquidCrystal.h>
const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
void setup() {
  // put your setup code here, to run once:
  pinMode(TEMP,INPUT);
  pinMode(MOTOR,OUTPUT);
  lcd.begin(16,2);
}
float in_v_temp;
int level_temp ;
void loop() {
  // put your main code here, to run repeatedly:
  level_temp = analogRead(TEMP);
  in_v_temp= level_temp * ( 5.0/1024 );
  lcd.print(in_v_temp*100);
  if (in_v_temp*100>=30){
    analogWrite(MOTOR,255);
    lcd.print("DONE1");
  }
  else if((in_v_temp*100>=25)&&(in_v_temp*100<30)){
    analogWrite(MOTOR,128);
    lcd.print("DONE2");
  }
  else{
    analogWrite(MOTOR,63);
    lcd.print("DONE3");
  }
  delay(100);
  lcd.clear();
}
```

OUTPUT :

- If the temperature is more than 30 → the speed will be the max. speed .
- If the temperature is between 25 and 30 → the speed will be the half of max. speed .
- If the temp is less than 25 → the minimum speed will be on.

THANK YOU

FOR PAYING ATTENTION